Mars Habitat

Campaign 1 - Case 03 - Arcadia Planitia, Mars

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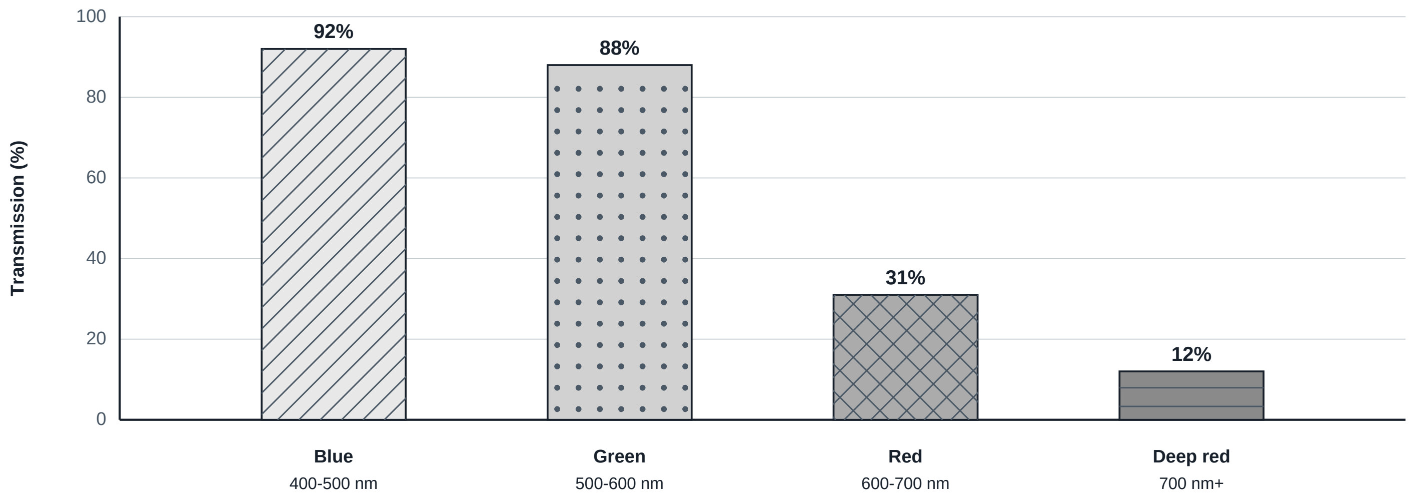
Completed exemplar answer key

TASK 1 - Define the measurement

PPFD tells how many photons in the defined photosynthetic waveband reach each square meter per second. It does not show how those photons are divided among blue, green, and red wavelengths.

TASK 2 - Read the spectral-transmission data

Surface intake to habitat output transmission



Blue transmission = 92%; green = 88%; red = 31%; deep red = 12%. Red is the lowest-transmission band inside 400-700 nm. Deep red is lower but is listed at 700 nm+.

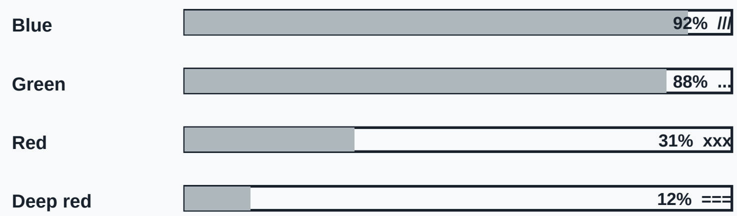
TASK 3 - Compare quantity and quality

Quantity check

280 umol m-2 s-1 combined PPFD

Runtime status: adequate total photon quantity

Spectral transmission by band



Analytical conclusion: adequate total does not prove an adequate wavelength distribution.

The 280 umol m-2 s-1 PPFD is described as adequate, so simple low total light conflicts with the sensor. The distribution is uneven: blue and green transmit 92% and 88%, while red and deep red transmit only 31% and 12%.

TASK 4 - Connect the symptom pattern

Older leaves already contain chlorophyll, while new leaves must make it. New tissue bleaching first therefore fits a failure in new chlorophyll formation better than root poisoning.

OLD LEAVES

Existing chlorophyll remains.

NEW LEAVES

New chlorophyll formation is disrupted.

ROOTS

Healthy roots weaken poisoning.

NUTRIENTS

Added iron and nitrogen did not help.

TASK 5 - Select and reject diagnoses

Select the wavelength-filtering diagnosis. Reject perchlorates because the roots are healthy and the crop uses a prepared nutrient system. CO_2 and photoperiod also do not fit the controlled evidence.

TASK 6 - Model the mechanism

47-sol filter replacement → **BP-4 rejects red/deep red** → **case-critical red delivery low** → **new chlorophyll development impaired** → **pale new growth**

Case mechanism model. General plant-light responses also involve blue, green, and red wavelengths.

Filter replaced 47 sols ago without a logged part number -> wrong BP-4 BLUE PASS filter -> blue/green remain high while red/deep red fall to 31%/12% transmission -> the game-defined light-dependent chlorophyll step is impaired -> pale or white new growth.

Acceptable alternatives: Equivalent wording is acceptable when it preserves selective attenuation, the case-specific POR/chlorophyll step, and new-tissue bleaching. Do not accept “plants only use red light.”

TASK 7 - Write the case conclusion

Claim: Wavelength-selective filtering, not low total light, causes the failure. Evidence: PPFD is adequate at 280; blue/green transmission is 92%/88%; red/deep-red transmission is 31%/12%; the filter was replaced 47 sols ago without a logged part number; new growth bleaches while older leaves retain green. Reasoning: PPFD combines photon quantity across wavelengths, so an adequate total can hide an unbalanced spectrum. The wrong BP-4 filter fits the selective pattern and the game-defined chlorophyll mechanism, matching the tissue pattern.

TASK 8 - Transfer the analysis

Increasing all wavelengths equally may raise PPFD without restoring the missing red proportion. First check the spectral distribution at canopy level, then correct the source or transmission path and remeasure.

TASK 9 - Exit ticket

Compare total PPFD with a wavelength-resolved spectral measurement. One measures total photon quantity; the other shows how that quantity is distributed.

Scoring note: Require accurate game values, a clear quantity-versus-spectrum distinction, a mechanistic link, and evidence-based alternative rejection. Game score and speed are excluded.